

HOOK_LOAD, TORQUE & WOB - SURFACE LOAD MONITORING

1. DATASET FIELDS REFERENCED IN THIS GUIDE

Field names below match the NWIS drilling_timeseries schema exactly, so this guide can be cross-referenced directly against logged data.

Field Name	Unit	Typical Range	Description
Hook_Load	klb	80 - 220 klb	Weight suspended from the traveling block/hook, including drillstring weight
Torque	kft-lb	8 - 22 kft-lb	Rotational resistance measured at surface
WOB	klb	8 - 25 klb	Weight on bit; derived as free-rotating Hook_Load minus on-bottom Hook_Load
RPM	rev/min	60 - 150 rpm	Rotational speed of the drillstring/bit

2. HOW THESE FIELDS RELATE TO EACH OTHER

Hook_Load vs WOB: WOB is calculated as the reduction in Hook_Load when weight is set down on the bit ($\text{WOB} = \text{free-rotating Hook_Load} - \text{on-bottom Hook_Load}$); a Hook_Load reading that does not drop as expected when WOB is applied suggests the bit is not actually taking weight (e.g. bridging or a washed-out interval above).

Hook_Load vs Overpull: Overpull is the amount by which Hook_Load while pulling out exceeds the free-rotating/static Hook_Load baseline; a rising Hook_Load trend during connections (with no corresponding WOB change) is the surface signature of increasing drag or an approaching stuck-pipe condition.

Torque vs Hook_Load (Combined Signature): Torque rising together with Hook_Load (overpull) indicates a mechanical pack-off; Hook_Load rising alone with Torque near baseline points instead toward differential sticking, since rotation is unaffected by a purely pressure-driven sticking mechanism.

RPM vs Torque under Stick-Slip: During stick-slip, downhole RPM cycles between near-zero and above-average even though the surface RPM setting is fixed, producing an erratic Torque signal that does not track a stable relationship with WOB.

3. KEY FORMULAS / RULES OF THUMB

Formula / Rule	Notes
$\text{WOB (klb)} = \text{Free-Rotating Hook_Load} - \text{On-Bottom Hook_Load}$	A quick data-quality check: WOB should never exceed the free-rotating Hook_Load reading.
$\text{Overpull (klb)} = \text{Hook_Load (while pulling)} - \text{Static/Free-Rotating Hook_Load baseline}$	Track over successive connections to build an escalation trend, not just a single reading.
$\text{Torque Baseline Ratio} = \text{Current Torque} / \text{Rolling 4-hr Average Torque}$	Ratios above 1.4-1.7x baseline are common tiered alarm thresholds (see Friction guide).

4. RISK INDICATORS & RECOMMENDED ACTIONS

Observed Pattern (Fields Involved)	Risk Indicated	Recommended Action
Hook_Load rising across connections, Torque stable	Differential sticking risk	Reduce Mud_Weight/ECD where pressure margin allows; prepare a freeing pill
Hook_Load AND Torque both rising across connections	Mechanical pack-off risk	Increase Flow_Rate/sweep frequency; plan a preventive wiper trip

WOB reading not decreasing when weight is set down (Hook_Load flat)	Bit not taking weight - possible bridge/ledge above bit	Work the string carefully back to bottom; do not force additional weight
Hook_Load spikes sharply with Torque simultaneously spiking	Sudden mechanical restriction (pack-off, whirl, or key seat)	Reduce WOB/RPM immediately and reassess before continuing

5. SAMPLE LOG ROWS (drilling_timeseries SCHEMA)

Well_ID	Timestamp	Depth_MD	Hook_Load	WOB	Torque	RPM
W002	2024-10-02 13:30	3160	165	0 (off bottom)	16.4	-
W005	2025-01-24 04:20	2740	142	10	22.0	70

6. FIELD_ALPHA CROSS-REFERENCE

- W002 (Alpha-02): Hook_Load-derived overpull escalated from 15 to 55 klb while Torque rose from 9 to 16 kft-lb across three connections before the string became immobile at Depth_MD 3160 m.
- W005 (Alpha-05): Torque spiked to 22.0 kft-lb at Depth_MD 2740 m while WOB was reduced from 14 to 10 klb as a mechanical response, illustrating the WOB-Torque adjustment used to break the event.

7. OPERATIONAL BEST PRACTICES

- Log Hook_Load continuously (not just at connections) so overpull trend detection has enough resolution to catch a gradual escalation.
- Always evaluate Hook_Load and Torque together to distinguish differential sticking from mechanical pack-off before choosing a response.
- Validate WOB against raw Hook_Load periodically to catch sensor drift or calibration issues.

This Drilling Parameter Reference Guide is a synthetically generated reference document produced for the NWIS prototype, aligned to the drilling_timeseries dataset schema. Not associated with an actual drilling operation.